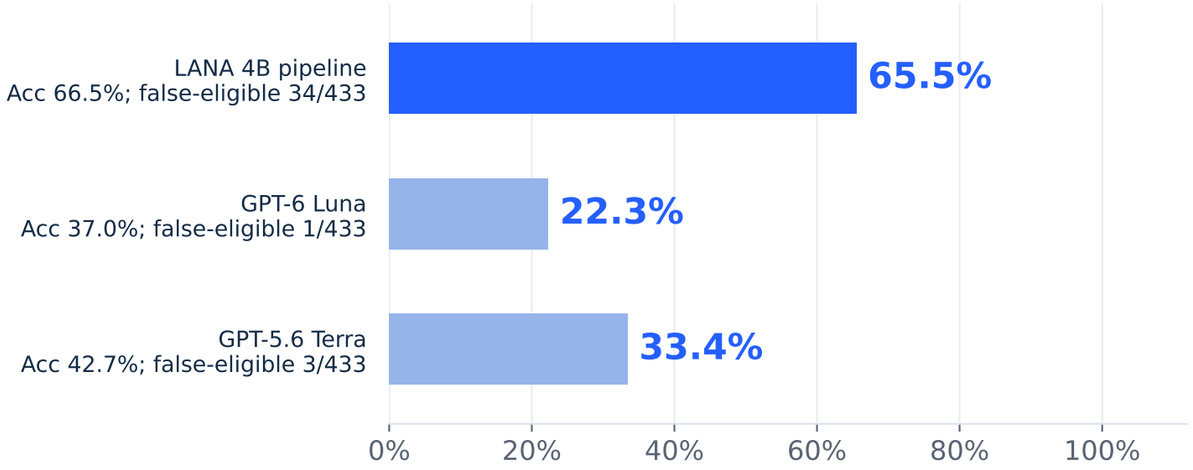
LANA CLINICAL-TRIAL MATCHING BENCHMARK

Local 4B pipeline vs Luna and Terra

Frozen confirmation: 665 pairs · 29 held-out synthetic profiles · 3 classes

QUALITY

Three-class macro-F1 (%) · Higher is better



Strict point wins on both metrics: yes · All four adjusted bounds above zero: yes

Adjusted LANA minus comparator intervals (percentage points; accuracy / macro-F1):

Luna: [+22.7, +35.8] / [+35.8, +50.0] · Terra: [+16.6, +30.6] / [+24.6, +39.5]

Eligible precision / recall: LANA: 73.8% / 41.4% · Luna: 80.0% / 1.7% · Terra: 57.1% / 1.7%

COST

Recorded USD / 1,000 decisions · Different scopes

LANA 4B pipeline — USD 0.015889

GPU electricity only · four uncached views

GPT-6 Luna — USD 0.188179

Recorded hosted API usage estimate

GPT-5.6 Terra — USD 3.078347

Recorded hosted API usage estimate

Whole-service local OPEX: not measured

One frozen adapted candidate and direct high-effort API controls. Paired patient-cluster bootstrap: 5,000 replicates; Bonferroni across four endpoints.
GPU electricity assumes USD 0.15/kWh; coverage 100.0%. API estimates include all attempts, not invoices. Cost scopes differ.
Host/CPU, idle, cold load, hardware, operations and research/training are outside local electricity; whole-service OPEX remains unmeasured.
Public synthetic general-medical challenge. No cancer, clinical-benefit, agentic-causality or outcome-RL claim. Earlier results retained.